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Principled Probability in Syntax: A Minimalist Approach to Variability and Cumulativity in Phrasal Movement

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1. Introduction

A primary goal of contemporary linguistic theory is to model humans' mental grammar; the system that exists in the mind of speakers that enables them to produce and comprehend language. A longstanding controversy involves how formal syntactic theories should approach language patterns where one set of features or grammatical relations can be expressed with more than one grammatical word order, with probabilistic variation. Traditionally, this debate has been dominated by two strongly opposed views, summarized in the next pagaraph (labels are my own).

On one view, mental grammars consist of PRINCIPLES WITHOUT PROBABILITY; Human language ability should be described purely in terms of structures and principles that apply categorically, without exception. Patterns that involve probabilistic variation (i.e. non-categorical outcomes) must be attributed to modules of the mind that are independent of the representation of language (Newmeyer 2003). To use a recent example in generative Minimalism, Titov (2020) posits that all syntactic movements apply categorically; optional movements must arise from operations in a postsyntactic module of grammar. The opposing view, PROBABILITY WITHOUT PRINCIPLES, claims that usage frequency, rather than abstract structures or principles, has a primary role in the representation of language in the human mind. In essence, speakers of a language attend directly to the frequency of linguistic units that they encounter, and the frequency of collocations between different units. Some works in this tradition further deny that mental grammars can contain abstract grammatical categories at all (Bybee and Hopper 2001).

This paper proposes that aspects of these two approaches should be reconciled to create an explanatory model of probabilistic word order patterns in grammar. Broadly, this is a theory of mental grammar with PRINCIPLES AND PROBABILITIES. I illustrate the approach with a case of word order variability in Cherokee (Iroquoian: Oklahoma and North Carolina), in which some features have probabilistic effects on clausal word order. I show that the pattern can be modeled within a Minimalist theory of syntactic representations and feature-triggered movement, integrated with a constraint-based, probabilistic model of grammatical computation, Maximum Entropy Grammar (Goldwater and Johnson 2003, Hayes and Wilson 2008). It is worth noting that the presented theory is concerned only with patterns that involve probabilistically variable outputs that emerge from one set of formal, syntactic features in the input; in other words, single Numerations that can produce multiple grammatical outputs. It is not concerned with variability across patterns that express distinct grammatical relations (Numerations that differ in content), nor with the frequency of occurrence of particular grammatical properties (the likelihood of a given Numeration).

2. Variability and cumulativity in Cherokee clausal word order

Cherokee allows a high degree of optionality in the placement of non-pronominal arguments (hence, NPs) relative to verbs. As shown in the following representative examples from Feeling et al. (2017:

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101, 43, 129, 35), all ordering permutations of agent NPs, theme NPs, and verbs are grammatical, and thematic arguments of a verb are not always realized as an NP.

(1) a. ᎩᏟ ᎣᎩᏍᏓᏩᏛᏒ
gitli ogi-sdawadvs-v
dog 1.PL.EXCL-follow-EXP
'The dog followed us.' [***Ag.>V***]

b. ᎠᎾᏓᏍᏕᎵᏍᎪ ᏴᏫ ᏧᎾᏍᏗ
a-n-adasdelis-g-o yvwi j-u-n-asdi
3A-PL-help-PROG-HAB people DST-3-PL-litle
'The little people help (others)' [***V>Ag.***]

(2) a. ᎠᎹ ᏥᏙᏗᏍᎪ
am ji-todis-g-o
water 1A-heat.water-PROG-HAB
'I heat water' [***Th.>V***]

b. ᎤᏍᏚᏅ ᎦᎶᎯᏍᏗ
u-sduhn-v galohisdiʔi
3B-close-EXP door
'(he) closed the door.' [***V>Th.***]

This section first summarizes the main findings of Hsu and Frey (2024) on word order variation in Cherokee, based on a corpus analysis of transcribed, spoken narratives from Montgomery-Anderson (2008) and Feeling et al. (2017). First, some aspects of clausal word order are (near) categorically predictable. Two types of items occur uniformly clause-initially, before all other items: *contrast-bearing expressions*, whose referent(s) belong to a contextually relevant set and are evoked to the exclusion of alternatives (Vallduví and Vilkuna 1998; Molnár 2002), and *frame-setting expressions*, which express the delimiting time or location of an event. The Hsu and Frey corpus has no clauses that contain both a contrasted expression and a frame-setter, so their relative order is indeterminate.

This paper focuses on the placement of non-contrasted argument NPs, which is variable. Generally, argument NPs can occur in either a *preverbal* or *postverbal* position. We can model the Cherokee clause as having two domains in which NPs can occur, as shown in (3). For convenience, I refer to the two relevant domains of the clause as CP and TP, remaining agnostic about their precise labeling.

(3) {Contrasted XP, Framesetting XP} > (XP) > Verb > (XP)
"CP" "TP"

Hsu and Frey (2024) find that the placement of non-contrasted argument NPs below CP is probabilistically determined by two properties, REFERENTIAL ACCESSIBILITY, a.k.a. "givenness" (Kuno 1972; Haviland and Clark 1974; Chafe 1976; Prince 1981) and THEMATIC ROLE (Jackendoff 1972; Gruber 1976). I briefly present the word-order preferences associated with each factor in turn.

Among NPs that refer to entities, *new* NPs that mention an entity for the first time in the discourse show the greatest likelihood of preverbal placement. At the other end of the scale, *given* NPs that have been explicitly mentioned in the same narrative show the weakest preference for preverbal placement. *Accessible* NPs are of an "intermediate" information status: their referents have not been directly evoked, but are inferrable either by a relationship with a given entity or from general knowledge. In terms of word order, accessible NPs show an intermediate preference for preverbal placement. Hsu and Frey classify NPs that do not refer to identifiable entities as *nonreferential*: this classification includes expressions that denote property or kind readings, or refer to events. Nonreferential and new NPs show a similarly high preference for preverbal placement.

(4) *Placement of NPs relative to verbs, by referential accessibility*

	Nonreferential	New	Accessible	Given
Percentage of preverbal placement	87% (67/77)	82% (90/110)	68% (53/78)	56% (128/229)

The word order effects of thematic roles are best observed in the contrast between *agents*, entities that initiate or cause the event denoted by the verbal predicate, and *themes*, entities that undergo a process, change of state, change of location, or continuation of state. Agent NPs show a greater propensity for preverbal placement than theme NPs, as shown in (5). This effect is not reducible to another factor, such as a subject-object asymmetry. As noted by Hsu and Frey, the word-order preferences of agents and themes is not sensitive to the number of arguments taken by the verb; Theme NPs of transitive verbs show the same ordering tendencies as theme arguments of unaccusative verbs.

(5) *Placement of agent and theme NPs relative to verbs*

	Agent	Theme
Percentage of preverbal placement	75% (65/87)	63% (145/232)

Crucially, the regression model in Hsu and Frey (2024) finds referential accessibility and thematic role to be *independent* factors in determining NP placement. The cumulative nature of these factors is seen in the cross pair table in (6). Across all levels of referential accessibility, agent NPs show a higher likelihood of preverbal placement than theme NPs (compare the left versus right column). NPs that denote new information show a higher likelihood of preverbal placement than given-information NPs, for both agent and and theme NPs (compare top, middle, and bottom rows within each column).

(6) Probability of preverbal placement, by thematic role and referential accessibility

	NP is an agent	*NP is a theme*
NP is new	**94%** preverbal (15/16)	**75%** preverbal (43/57)
NP is accessible	**77%** preverbal (10/13)	**70%** preverbal (28/40)
NP is given	**68%** preverbal (36/53)	**51%** preverbal (56/109)

The pattern suggests that some principles that determine NP placement in Cherokee apply in a probabilistic manner, while holding constant the formal features present in the numeration. It is important to note that this variability cannot be explained away as the result of categorical effects of other potential factors. The regression analysis in Hsu and Frey (2024) finds no significant word order effects arising from NP animacy, NP phonological length, or speaker identity. NP placement is also not predictable from the presence or absence of a verbal agreement prefix (Humphreys and Hsu 2025).

Furthermore, variability in NP placement cannot be due to the information-structural categories of focus or topic. The results above are not likely skewed by effects of focus, given that the corpus consists only of narratives produced by a single speaker, in which the vast majority of sentences occur in a broad focus context (answering the implicit question "what happened next?"). I have not found sentences with a clear partition of presupposed versus focused domains. The corpus contains several NPs that can reliably be identified as topics: overt NPs that correspond with pro-dropped arguments in one or more immediately following clauses. This includes *continuing topics*, where the NP is also an argument of the preceding clause, and *shifted topics*, where the NP is not an argument of the preceding clause (Frascarelli and Hinterhölzl 2007). Each types of topic is attested in both preverbal and postverbal positions. The first sentence in example (7) shows a preverbal shifted topic (*that old brown dog of ours*), and its last sentence shows a preverbal continuing topic (*that dog*). Each of these corresponds to a pro-dropped argument in the next clause, some of which are omitted here for brevity.

(7) Context: The narrator, his family, and their dog have been at a neighbor's house. They have been preparing to leave.

a.

Ꮎ	**ᎤᏬᏗᎨ**	**ᎩᏟ**	**ᎣᎩᎧᎲ**	ᏚᎴᏅ
Na	**u-wodige**	**gitli**	**ogi-kah-v**	d-u-len-v.
that	**3B-brown**	**dog**	**1.PL.EXCL-have.ANI.OBJ-EXP**	DST-3B-get.up-EXP

'the brown dog of ours got up.'

b.

ᏙᏱ	ᏗᎸ	ᎠᏲᏓᏢᎲ	ᎦᏅᎬᎢ.
doyi	dilv	ayodatlahv	ga-nv-g-vʔi.
outside	toward	porch	3A-lie-PROG-EXP

'He was lying outside on the porch.'

c.

ᎥᏍᎩᏃ	**ᎩᏟ**	ᎣᎨᎳᏛᏒ	...
vsgi=no	**gitli**	ok-eladvs-v	
that=CN	**dog**	1.PL.EXCL-follow-EXP	

'That dog followed us ...' (Feeling et al. 2017: 102)

Example (8) shows a postverbal shifted topic (*the men*), and (9) has a postverbal continuing topic.

(8) Context: Two men in a canoe have just seen that a bull has swum underneath them.

ᏚᏟᏆᏗᏁᎴ	ᏥᏳ.	ᏚᏂᎬᏤᏃ	**ᎠᏂᏍᎦᏯ**
d-u-hlihgwadinel-e	jiyu.	d-u-ni-gvj-e=hno	**a-ni-sgaya**.
PL-3B-turn.over-REPP	canoe	DST-3B-PL-fall.in-REPP=CN	**3A-PL-man**
'The canoe turned over.'		The men fell in the water.'	(Feeling et al. 2017: 111)

(9) Context: The narrator is describing a group of dogs who have been bothered by a cat's meowing.

ᎠᎾᏁᎷᎩᏍᎪ	ᎦᎵᏦᏕ.	ᏭᏂᏕᏯᏍᏗᎮ	**ᎩᏟ**
a-n-anelugis-g-o	galjode.	w-u-ni-deysddih-e	**gitli**
3-PL-run.after-PROG-HAB	house	TR-3-PL-run.around-REPP	**dog**
'They would chase it around the house.		The dogs would run around.'	(Feeling et al. 2017: 81)

This suggests that topichood (or subtypes thereof) is not a categorical word order predictor in Cherokee, even though it may well turn out to have probabilistic effects.

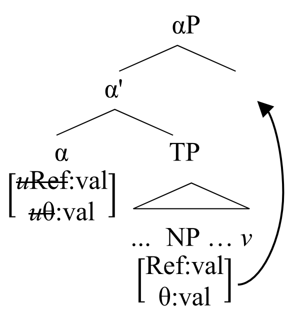
Finally, postverbal NPs cannot be treated as parenthetical expressions or afterthoughts that are weakly integrated with the clause, which would allow an analysis of the Cherokee clause as uniformly verb-final. Such an analysis is not generally tenable because postverbal NPs can introduce new entities to the discourse (see table in (4)), which would be highly uncharacteristic of a parenthetical expression.

3. Agreement and movement

I assume that syntactic representations contain information-structural features, which can participate in Agree relations (Rizzi 1997; Miyagawa 2009; Ostrove 2018; Kratzer and Selkirk 2020; Mursell 2021).

We restrict our attention to the portion of the Cherokee clause below CP (which hosts framesetting and contrast-bearing NPs). I propose that preverbal NPs occur within TP, and that verbs occur at the end of TP. Postverbal NP placement results from agreement between a preverbal NP and a head α that occurs between T and C (Miyagawa 2009), followed by NP Movement to a rightward specifier of αP. For similar analyses of givenness-sensitive rightward movement, see Diercks (2022) and Bobaljik (2025).

(10)



Turning to the features relevant to agreement, I assume that NPs bear valued features [θ:val] and [Ref:val], corresponding to their thematic role (Lasnik 1995; Hornstein 1998; Sabbagh 2025) and referential accessibility (Kratzer and Selkirk 2020; Mursell 2021). The head α contains two corresponding probes, [*u*θ:] and [*u*Ref:], valued by Agree with the same NP.[1] The probability of subsequent phrasal movement to the specifier of αP depends on the values obtained by each probe on α. The precise implementation of this is given in the following constraint-based analysis.

The analysis of postverbal NP placement as being generated by rightward movement captures several aspects of the Cherokee pattern that prove more challenging to an alternative analysis in which

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[1] While Kumaran (2022) argues that constraint interaction itself determines whether Agree occurs, I assume that the operation always occurs in Cherokee. Verbal agreement morphology does not apparently interact with NP placement (Humphreys and Hsu 2025).

preverbal NP placement is generated by leftward movement from a postverbal base position. First, the proposed analysis better accounts for the observation that Cherokee sentences are overwhelmingly either verb-final, or have exactly one postverbal XP (98% of clauses in the Hsu and Frey corpus). This restriction on postverbal constituents is predicted to arise if αP is the sole projection that permits rightward specifiers. In contrast, the alternative leftward movement analysis leaves little way to explain why leftward movements to a preverbal position are generally optional, but obligatory if non-movement would leave more than one XP after the verb. Second, the proposed analysis is consistent with the observation by Kratzer and Selkirk's (2020) that [given] patterns like an active feature in ways that [new] does not. On the rightward movement analysis, the pressure to move an NP is proportional to its degree of givenness. Finally, proposal concurs with the insight of Bobaljik (2025) that non-referential arguments are exempt from the referential accessibility scale, and thus expected to be immune to movement operations that depend on referential accessitility. On this view, the near-categorical preverbal placement of non-referential NPs in Cherokee may arise because [*u*Ref:] cannot agree with a non-referential NP.

4. Violable constraints and gradient cumulativity

Probabilistic and *cumulative* patterns of movement triggering challenge the view that the ability to trigger movement is inherent to heads or probes (Chomsky 1993). An approach where the movement-triggering property applies obligatorily, but is optionally present on some heads (Chomsky 2000), leaves no explanation for when it occurs (Titov 2020), nor why it would correlate with feature values involved in agreement.[2] However, probabilistic and cumulative patterns can be modeled in models of grammatical computation that use numerically weighted, violable constraints, as shown in many works in generative phonology (Zuraw and Hayes 2017; Smith and Pater 2020). Here, I present an analysis of the Cherokee pattern in a theory that integrates Minimalist representations with constraint-based optimization: Harmonic Minimalism (Heck and Müller 2003, 2013; Murphy 2017; Hsu 2021; Müller et al. 2022).

In this framework, as in other varieties of Minimalism, syntactic structures are built derivationally from the bottom up (Chomsky 1993). At each derivational step, the grammar examines the built structure and compares output candidates that apply different operations (Heck and Müller 2013). In order to capture the probabilistic nature of output selection observed in Cherokee, I adopt Maximum Entropy Harmonic Grammar (MaxEnt: Goldwater and Johnson 2003; Hayes and Wilson 2008) for the computation of optimality. All constraints are associated with a numerical *weight* (Legendre et al. 1990), rather than being subject to a ranking (Prince and Smolensky 1993). Rather than selecting a single optimal candidate, the grammar generates a probability distribution over output candidates, calculated from the harmony scores of each candidate. Candidates with lower harmony are less likely to surface than more well-formed candidates, but are not categorically banned.

We now turn to the relevant set of output candidates and constraints. We focus on the derivational step that occurs once α has been Merged, and both [*u*θ:] and [*u*Ref:] have agreed with an NP. At this step, the grammar generates two types of output candidates: one in which the NP moves to the specifier of αP, and one in which no movement occurs. Following Müller (2019), I assume that any movement violates the ANTI-LOCALITY CONDITION constraint, defined in (11). In Müller's implementation, the degree of violation depends on which which heads are crossed. As all movements to Spec, αP cross the same maximal constituent TP, I assume as a simplification that each movement incurs one violation of this constraint. Agreement that does not result in movement violates a FEATURE CONDITION constraint (Heck and Müller 2003), defined in (12).

(11) ANTI-LOCALITY CONDITION: For all heads Y, *Y that c-commands α_i of dependency δ and m-commands α_{i-1} of δ.

[2] Titov posits that variable movement patterns emerge in a postsyntactic module, arising from constraints on the correspondence between LF and PF representations. While such an approach is plausible with information-structural features, which often have prosodic correlates, it is less plausible to posit constraints that refer to the linearization of XPs based on their thematic roles, as would be required to model the Cherokee pattern.

(12) FEATURE CONDITION: For each probe [uF] that agrees with an XP with matching [F], XP occurs in the specifier of the head that contains the probe.

Following Hsu (2021), I posit that the grammar contains multiple instantiations of the latter constraint, corresponding to each distinct probing feature. This permits grammars in which FEATURE CONDITION constraints differ in weight, such that probes to differ in their propensity to trigger movement. After both probes have agreed with an NP, the grammar compares output candidates with phrasal movement, which violate ANTI-LOCALITY, and those without movement, which violate FEATURE CONDITION constraints.

(13) *Basic violation profiles of output candidates with movement versus non-movement*

$[_{\alpha'}$ α $\begin{bmatrix} u\text{Ref:val} \\ u\theta\text{:val} \end{bmatrix}$ $[_{TP}$ … NP $\begin{bmatrix} \text{Ref:val} \\ \theta\text{:val} \end{bmatrix}$ ... V]	FEATURE CONDITION (REF)	FEATURE CONDITION (θ)	ANTI-LOCALITY
$[_{\alpha P}$ $[_{\alpha'}$ α $[_{TP}$ … ~~NP~~ ... V] NP]			-1
$[_{\alpha P}$ α $[_{TP}$ … NP ... V]]	-1	-1	

FEATURE CONDITION constraints are defined such that no violation occurs if there is no potential goal for that probe in the structure. I assume that FEATURE CONDITION (REF) is not violated if there is no [REF:] feature that can participate in Agree, which occurs if all NPs are nonreferential.

Finally, the Cherokee pattern requires FEATURE CONDITION penalties to be sensitive to the feature values involved in agreement, because these distinctions influence the probability of phrasal movement. Furthermore, the grammar should capture the implicational relationship between an NP's degree of referential accessibility and its likelihood of postverbal placement: given > accessible > new. We do not expect to find a language, for instance, where accessible NPs show a greater (or weaker) preference for movement than both given and new NPs. To account for this, I posit that FEATURE CONDITION violations are *scaled* according to the prominence of feature values obtained via agreement.[3] In particular, each instance of agreement without movement incurs a FEATURE CONDITION penalty of $w + (s \times d)$, where:

(14) w = the basic weight of the constraint
s = a constraint-specific scaling factor
d = a value along a prominence scale (0, 1, … n), determined by a feature value in the Agree relation

I use the following prominence scales for each FEATURE CONDITION constraint:

(15) d for FC(REF): [Ref: new] = 0, [Ref: acc] = 1, [Ref: given] = 2
d for FC(θ): [θ: agent] = 0, [θ: theme] = 1

Given these scales, as long as s is greater than zero, failing to move a more discourse-given NP incurs a greater penalty of FEATURE CONDITION (REF) than failing to move a discourse-new one, and failing to move a theme NP incurs a greater penalty of FEATURE CONDITION (θ) than failing to move an agent NP.

A set of constraint weights that generates the attested pattern is identified using the MaxEnt Grammar Tool learner (Wilson and George 2009). The learner is supplied with tableaux including (i) candidate output forms, (ii) their violation profiles, and (iii) their frequencies, corresponding to the cross-pair table in Section 2. The constraint weights acquired by the learner on the basis of this input are shown in the tableaux below. Each tableau shows the learned constraint weights (w) and the harmony scores (H), and predicted probabilities (P) associated with particular output candidates.

[3] For uses of constraint scaling in generative phonology, see Flemming (2001) and overviews in McPherson and Hayes (2016); Hsu and Jesney (2018). See Georgi (2017) for an application of constraint scaling to cumulative prominence-scale effects in verbal agreement.

The three tableaux below illustrate the analysis for agent NPs of each referential accessibility value, corresponding to the left column of the cross-pair table in Section 2. Each tableau differs only in the total penalty of the FEATURE CONDITION (REF) violation, determined by the scaling factor *s*=0.6. Each successive increase in the referential accessibility value of the NP results in a lower total harmony of the candidate without movement, decreasing the likelihood of preverbal placement.

(16) *Clause with new-information agent NP*

$[_{\alpha'}\ \underset{\begin{bmatrix}\cancel{u\mathrm{Ref}}\text{:new}\\ \cancel{u\theta}\text{:agent}\end{bmatrix}}{\alpha}\ [_{TP} \ldots \underset{\begin{bmatrix}\mathrm{Ref}\text{:new}\\ \theta\text{:agent}\end{bmatrix}}{\mathrm{NP}} \ldots \mathrm{V}]$	FC(REF) *w*=1 *s*=0.6	FC(θ) *w*=1 *s*=0.73	AL *w*=4	*H*	***P***
$[_{\alpha P}\ [_{\alpha'}\ \alpha\ [_{TP} \ldots \cancel{\mathrm{NP}} \ldots \mathrm{V}]\ \mathrm{NP}]$			-1	-4	**.12**
$[_{\alpha P}\ \alpha\ [_{TP} \ldots \mathrm{NP} \ldots \mathrm{V}]\ ]$	-1_{new} =1+(.6×0)	-1_{agent} =1+(.73×0)		-2	**.88**

(17) *Clause with accessible agent NP*

$[_{\alpha'}\ \underset{\begin{bmatrix}\cancel{u\mathrm{Ref}}\text{:acc}\\ \cancel{u\theta}\text{:agent}\end{bmatrix}}{\alpha}\ [_{TP} \ldots \underset{\begin{bmatrix}\mathrm{Ref}\text{:acc}\\ \theta\text{:agent}\end{bmatrix}}{\mathrm{NP}} \ldots \mathrm{V}]$	FC(REF) *w*=1 *s*=0.6	FC(θ) *w*=1 *s*=0.73	AL *w*=4	*H*	***P***
$[_{\alpha P}\ [_{\alpha'}\ \alpha\ [_{TP} \ldots \cancel{\mathrm{NP}} \ldots \mathrm{V}]\ \mathrm{NP}]$			-1	-4	**.20**
$[_{\alpha P}\ \alpha\ [_{TP} \ldots \mathrm{NP} \ldots \mathrm{V}]\ ]$	-1_{acc} =1+(.6×1)	-1_{agent} =1+(.73×0)		-2.6	**.80**

(18) *Clause with given agent NP*

$[_{\alpha'}\ \underset{\begin{bmatrix}\cancel{u\mathrm{Ref}}\text{:giv}\\ \cancel{u\theta}\text{:agent}\end{bmatrix}}{\alpha}\ [_{TP} \ldots \underset{\begin{bmatrix}\mathrm{Ref}\text{:giv}\\ \theta\text{:agent}\end{bmatrix}}{\mathrm{NP}} \ldots \mathrm{V}]$	FC(REF) *w*=1 *s*=0.6	FC(θ) *w*=1 *s*=0.73	AL *w*=4	*H*	***P***
$[_{\alpha P}\ [_{\alpha'}\ \alpha\ [_{TP} \ldots \cancel{\mathrm{NP}} \ldots \mathrm{V}]\ \mathrm{NP}]$			-1	-4	**.31**
$[_{\alpha P}\ \alpha\ [_{TP} \ldots \mathrm{NP} \ldots \mathrm{V}]\ ]$	-1_{giv} =1+(.6×2)	-1_{agent} =1+(.73×0)		-3.2	**.69**

The next three tableaux illustrate the analysis for theme NPs of each referential accessibility value, corresponding to the right column of the cross-pair table. They differ from the tableaux above only in having a greater scaled penalty of FEATURE CONDITION (θ), resulting in a lower harmony for all candidates without movement, as compared with each of the three tableaux above.

(19) *Clause with new-information theme NP*

$[_{\alpha'}\ \underset{\begin{bmatrix}\cancel{u\mathrm{Ref}}\text{:new}\\ \cancel{u\theta}\text{:theme}\end{bmatrix}}{\alpha}\ [_{TP} \ldots \underset{\begin{bmatrix}\mathrm{Ref}\text{:new}\\ \theta\text{:theme}\end{bmatrix}}{\mathrm{NP}} \ldots \mathrm{V}]$	FC(REF) *w*=1 *s*=0.6	FC(θ) *w*=1 *s*=0.73	AL *w*=4	*H*	***P***
$[_{\alpha P}\ [_{\alpha'}\ \alpha\ [_{TP} \ldots \cancel{\mathrm{NP}} \ldots \mathrm{V}]\ \mathrm{NP}]$			-1	-4	**.22**
$[_{\alpha P}\ \alpha\ [_{TP} \ldots \mathrm{NP} \ldots \mathrm{V}]\ ]$	-1_{new} =1+(.6×0)	-1_{theme} =1+(.73×1)		-2.73	**.78**

(20) *Clause with accessible theme NP*

$[_{\alpha'}$ α $[_{TP}$ … NP … V] [~~*u*Ref~~:acc, ~~*u*θ~~:theme] [Ref:acc, θ:theme]	FC(REF) *w*=1 *s*=0.6	FC(θ) *w*=1 *s*=0.73	AL *w*=4	*H*	***P***
$[_{\alpha P}$ $[_{\alpha'}$ α $[_{TP}$ … ~~NP~~ … V] NP]			-1	-4	**.34**
$[_{\alpha P}$ α $[_{TP}$ … NP … V]]	-1_{acc} =1+(.6×1)	-1_{theme} =1+(.73×1)		-3.33	**.55**

(21) *Clause with given theme NP*

$[_{\alpha'}$ α $[_{TP}$ … NP … V] [~~*u*Ref~~:giv, ~~*u*θ~~:theme] [Ref:giv, θ:theme]	FC(REF) *w*=1 *s*=0.6	FC(θ) *w*=1 *s*=0.73	AL *w*=4	*H*	***P***
$[_{\alpha P}$ $[_{\alpha'}$ α $[_{TP}$ … ~~NP~~ … V] NP]			-1	-4	**.48**
$[_{\alpha P}$ α $[_{TP}$ … NP … V]]	-1_{giv} =1+(.6×2)	-1_{theme} =1+(.73×1)		-3.93	**.52**

Given the previous claim that nonreferential NPs lack a valued [Ref:] feature, and thus incur no violation of FEATURE CONDITION (REF), this grammar predicts that nonreferential NPs will have a greater propensity to remain in a preverbal position than discourse-new NPs. While this is true of the total counts in the Hsu and Frey corpus (see example (4), the distinction between nonreferential and new NPs is not statistically significant in their regression analysis, so the conclusion remains tentative.

A natural question arises on how the approach extends to clauses that contain two argument NPs. Although such clauses are uncommon in naturalistic speech, it is possible for clauses to have two postverbal argument NPs (Hsu and Frey 2024; e146–e147). I tentatively suggest that the probes on α agree with each argument NPs in the clause, and that the movement of each NP is determined by the constraint system shown above. I leave a comprehensive evaluation of this prediction to future work.

It is worth noting that the effect of an NPs feature values on its propensity to move could also be modeled in Gradient Harmonic Grammar (Smolensky and Goldrick 2016) by positing that feature values influence the degree of activity of NPs (Müller 2019; Müller et al. 2022; Amato 2024), proportionally altering constraint penalties. The main empirical difference between such an approach and the FEATURE CONDITION scaling used in this paper is that adjustments to the gradient activity of NPs are predicted to affect the penalties of *all* constraints that refer to them, beyond the relevant FEATURE CONDITION constraints. I do not at present know of patterns in Cherokee that lend unique support to either alternative.

5. Conclusion

This paper has presented an approach to formal syntax characterized by PRINCIPLES AND PROBABILITIES. This model affirms the value of discoveries about grammar made by formal theories that assume full categoricity, while expanding their power to explain variable phenomena observed in many domains of natural language. Specifically, I have shown that the representational mechanisms of Minimalist syntactic theory (Chomsky, 1993, 2000, among many others), when paired with a constraint-based model of grammatical computation, can successfully model probabilistic phrasal movement patterns, as exemplified by clausal word order in Cherokee. The approach preserves the insight of Minimalism that movement is *feature-driven*, while predicting that some movements may apply probabilistically.

I conclude by noting several broader implications of the analysis. First, it challenges the assumption that different syntactic structures or linear orders necessarily express distinct meanings (Newmeyer 2003: 697), as also argued recently by Schoenmakers et al. (2021) on the basis of object scrambling in Dutch. Second, the study adds to findings that word order preferences can depend on cumulative interactions of more than one syntactic property, in both probabilistic patterns (Ellsiepen and Bader

2018) and categorical ones (Murphy 2018). Third, although it is sometimes assumed that variable movements can only be triggered by information-structural features, the Cherokee pattern suggests that other properties, such as thematic roles, can influence probabilistic movements (see also Cable 2012 for a case of optional movement following φ-feature agreement). Broadly, I hope to have shown that the investigation of probabilistic word order patterns provides fruitful avenues of research in generative syntax, with great potential significance for theories of clause structure, syntactic agreement, and the featural triggers of word order variation.

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